

Designing Data-Intensive Applications by Martin Kleppmann

Designing Data-Intensive Applications explains how to build reliable, scalable, and maintainable software systems that handle large amounts of data. The book covers databases, distributed systems, storage, data processing, and system design trade-offs.

Main Ideas:

- **Reliability:** Systems should continue working despite failures.
- **Scalability:** Applications must handle growth in users and data.
- **Maintainability:** Systems should stay understandable and easy to improve.
- **Data Models:** Different databases fit different needs.
- **Distributed Systems:** Networks create challenges like replication and consistency.
- **Trade-Offs:** Engineering choices always involve benefits and costs.

Overall Message:

Strong software systems come from understanding data, failures, and careful design decisions.